INCH-POUND

MIL-F-82795(OS)
29 December 1989
SUPERSEDING
WS 20621
25 October 1983

MILITARY SPECIFICATION

FLOATS, PULP, ASBESTOS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an asbestos pulp float used in rocket motor and igniter liners (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 11 Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

CHRYSOTILE ASBESTOS TEST MANUAL (Third Edition)

B-2-72 Ro-Tap Screen Analysis
D-1-72 Wet Volume
G-2-74 Quantitative Analysis
G-4-73 Moisture Content

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(Application for copies should be addressed to the Quebec Asbestos Mining Association, 580 Grande Allee East, Suite 320, Quebec, P.Q. Canada, G1R 2K2.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Material.** The floats shall be natural asbestos fibers suitable for use as a filler in adhesive compositions.

3.2 **Chemical and physical properties.** Chemical and physical properties shall be as specified in table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Volatile matter, percent by	--	1.0
pH (aqueous extract)	8.8	9.8
Calcination loss, percent by weight	--	16.8
Fiber size distribution passing through number 10 sieve, percent	100	--
Wet volume in 50-g sample, milliliters	250	600

3.3 **Safety.**

3.3.1 **Toxic products.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

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3.4 **Shelf life.** The floats shall have a shelf life of 36 months after date of manufacture when stored at ambient temperature in the original closed container.

3.4.1 **Shelf life extension.** The shelf life expiration date of an individual lot may be extended for additional 36-month periods provided the material, upon retest, conforms to the volatile matter requirement specified in table I.

3.5 **Workmanship.** Workmanship shall be such that the asbestos pulp floats are uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

4.2.1 **Lot.** A lot shall consist of the asbestos pulp floats subjected to inspection at one time from one of the following:

a. A single batch (see 6.3.1).

b. A continuous batch (see 6.3.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

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TABLE II. Quality conformance inspections.

Test	Requirement	Test Method
Visual examination	3.5 and Section 5	4.3.1
Volatile matter	3.2	4.3.2
pH (acqueous solution)	3.2	4.3.3
Calcination loss	3.2	4.3.4
Fiber size distribution	3.2	4.3.5
Wet volume	3.2	4.3.6

4.2.1.1 **Lot increments.** When a lot is delivered in increments, each increment may be treated as a lot for acceptance purposes. Acceptance of an incremental shipment may be based on data obtained from tests on lot samples, tests on incremental shipment samples, or from a combination of both.

4.2.2 **Sampling.** Random samples shall be taken from each lot. When a lot is delivered in increments, random samples may be taken from each increment. Each sample and the records thereof shall be identified as to material, lot number, container number, and the samples shall be prepared and evaluated as specified herein. Where samples are to be taken from the shipping containers, a table of random numbers shall be used for selecting those shipping containers from which the samples shall be removed. The number of containers to be sampled shall be based on the total number of containers in the lot or increment and as specified in table III. When stream sampling is used, the number of samples to be taken shall be equal to the number which would be taken if the containers were sampled. The stream samples shall be taken while filling those shipping containers selected for sampling using a table of random numbers and as specified in table III.

4.2.2.1 **Sample preparation.** The single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites except where a lot consists of fewer than three containers. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of less than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended in accordance with table III. The results of each test shall meet the requirements of table I. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

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TABLE III. Sampling.

No. of containers per lot	No. of containers sampled per lot	Single samples per composite		
		Composite 1	Composite 2	Composite 3
1	1	1	1	1
2	2	2	2	2
3 - 15	3	1	1	1
16 - 25	4	1	1	2
26 - 40	5	1	2	2
41 - 65	7	2	2	3
66 - 110	10	3	3	4
111 - 180	15	5	5	5
181 - 300	20	6	7	7
301 - 500	25	8	8	9
501 - 800	30	10	10	10
801 - 1300	35	11	12	12

WARNING: Because of potential danger, precautions should be taken to protect personnel from hazardous irritants. It is mandatory that goggles and dust respirators be used when handling asbestos pulp floats.

4.3 **Test methods.** The following test methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.3.1 **Visual examination.** Visual examination shall be made to verify conformance to 3.5 and Section 5.

4.3.2 **Volatile matter test.** Volatile matter content shall be determined in accordance with the following:

a. Weigh a 4- to 5-gram sample to the nearest milligram (mg) and place in a tared low-form glass-stoppered weighing bottle.

b. Heat in an oven at 105 + 5 - 0°C for 60 ± 5 minutes with the stopper slightly ajar.

c. Remove from the oven, cool to room temperature in a desiccator, and weigh to the nearest mg.

d. Calculate volatile matter lost using the following formula:

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$$\text{Percent volatile matter} = \frac{100\ (A - B)}{W}$$

Where A = weight of weighing bottle and sample, g

B = weight of weighing bottle and dried sample, g

W = sample weight, g

e. Record volatile matter lost to the nearest 0.1 percent

f. An alternative method for determining volatile matter content shall be in accordance with Method G-4-73 of the Chrysotile Asbestos Test Manual with values recorded to the nearest 0.1 percent.

4.3.3 **Hydrogen ion (pH) test.** pH shall be determined in accordance with the following:

a. Weigh a 4-g sample to the nearest 0.1 g.

b. Place the sample in a beaker containing 10 milliliters (ml) of freshly boiled distilled water and 2 ml of ethanol (95 percent).

c. Add 70 ml of boiling distilled water, cover the beaker with a watch glass and boil for 15 minutes.

d. Cool to room temperature, decant, and measure the pH of the supernatant liquid with a pH meter capable of 0.1 pH unit accuracy and previously calibrated with standard buffer solutions.

e. Record the pH to the nearest 0.1 unit.

4.3.4 **Calcination loss test.** Calcination loss shall be determined in accordance with the following:

a. Weigh a 4-g sample to the nearest 0.1 mg and place in a tared platinum crucible, with lid, which has been heated to 900 $\pm$ 50°C for 60 $\pm$ 5 minutes and cooled to room temperature in a desiccator.

b. Heat the crucible for 4 hours $\pm$ 15 minutes at 900 $\pm$ 50°C and cool to room temperature in a desiccator.

c. Weigh the crucible with ignited samples.

d. Calculate the calcination loss by the following formula:

$$\text{Percent calcination loss} = \frac{100\ (A - B)}{W}$$

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Where A = Weight of crucible and sample, g

B = Weight of crucible and ignited sample, g

W = Sample weight, g

e. Record calcination loss to the nearest 0.1 percent.

f. An alternative method for determining calcination loss shall be in accordance with Method G-2-74 of the Chrysotile Asbestos Test Manual.

4.3.5 **Fiber size distribution.** Fiber size distribution shall be determined in accordance with the following:

a. Weigh a 20-g sample to the nearest 0.1 g.

b. Place the sample, with 0.2 g of tricalcium phosphate, into a bottle and shake for 1 minute.

c. Sieve the treated sample for a minimum of 13 minutes on a sieve shaker using a U.S. Standard No. 10 (ASTM E 11) sieve for the top sieve and the required number of other sieves and a pan to fill the machine.

d. Visually examine the No. 10 sieve for retained fibers.

e. Record conformance or nonconformance to table I.

f. An alternative method for determining fiber size distribution shall be in accordance with Method B-2-72 of the Chrysotile Asbestos Test Manual. Results shall be recorded stating conformance or nonconformance to table I.

4.3.6 **Wet volume test.** Wet volume shall be determined in accordance with the following:

a. Weigh a 50-g sample to the nearest 0.05 g.

b. Add the sample to 1000 $\pm$ 200 ml of distilled water in a 2000-ml capacity glass cylinder that has an 80-millimeter diameter and is graduated into 20-ml subdivisions.

c. Add distilled water to bring the volume to 2000 ml and seal the top to prevent leakage.

d. Place the sealed cylinder in the fixture, rotate for 30 revolutions, and let stand vertically for 10 minutes.

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e. Rotate the cylinder an additional 30 revolutions, remove from the fixture, and place the cylinder in the vertical position, seal end up, on a vibration-free table. Allow the fibers to settle for 4 hours ± 15 minutes.

f. Visually determine the wet volume of the settled fibers in the cylinder.

g. Record the wet volume to the nearest 10 ml.

h. An alternative method for determining wet volume shall be in accordance with Method D-1-72 of the Chrysotile Asbestos Test Manual. Results shall be recorded to the nearest 10 ml for a 50-g sample.

4.4 **Inspection of packaging.** The float containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The asbestos pulp floats shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Net weight of contents.

d. Shelf life and inspection/test date for extension (see 3.4.1). In addition, the marking shall include the following:

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"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR ADDITIONAL 36-MONTH PERIODS PROVIDED THE FLOAT,UPON RETEST, CONFORMS TO THE VOLATILE MATTER REQUIREMENT OF MIL-C-82795."

e. Storage conditions.

f. "CAUTION" labels in accordance with 49 CFR Part 1910, Section 1910.93a(g) (2), shall be affixed to all raw material containers containing asbestos fibers. The "CAUTION" labels required shall be printed in letters of sufficient size and contrast to be readily visible and legible. The label shall state:

"CONTAINS ASBESTOS FIBERS AVOID CREATING DUST BREATHING ASBESTOS DUST MAY CAUSE SERIOUS HARM"

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The asbestos pulp floats specified herein are intended for use in the Standard Missile Mk 104 Dual Thrust Rocket Motor and igniter liners.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. Material Safety Data Sheet required (see 3.3.1 and 6.5).

d. Special marking, if other than as specified in 5.2.2.

e. Safety precautions required (see 6.6).

6.3 Definitions.

6.3.1 Batch. A batch is that amount of asbestos pulp floats made as one unit in the manufacturing process.

6.3.2 Continuous process. A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the material is accumulated for a lot.

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6.4 **Storage.** After receipt of material, the procuring activity is responsible for storage and retesting to determine its shelf life extension.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N800)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-82795	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
FLOATS, PULP, ASBESTOS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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